Supplementary Information to Quantifying Uncertainty of Predictions from Cancer Progression Models

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S1 Metric Tensor for smMALA

As suggested by Girolami and Calderhead (2011), we use the Fisher information matrix minus the Hessian of the log-prior, eq. (7), as a metric tensor for smMALA.

S1.1 Computing and Using the FIM for MHN

For the computation of the Fisher information matrix, we built upon Vocht (2022)'s implementation of the Fisher information matrix for cMHN. We improved the runtime, added a GPU implementation and generalized it to oMHN.

The implementations can be found in *mhn/full_state_space/fisher.pyx* and *mhn/full_state_space/cuda_full_state_space.cu* in the `mhn` repository in version 1.2.0.

S1.1.1 The FIM for oMHN

The FIM for oMHN can directly be computed from the FIM of its corresponding cMHN: For every oMHN, there is a cMHN that describes the same probability distribution: The function

$$\begin{aligned} c : \mathbb{R}^{(n+1)\times n} &\to \mathbb{R}^{n\times n} \\ \begin{pmatrix} (\theta_{ij})_{ij} \\ (\omega_j)_j \end{pmatrix} &\mapsto \begin{pmatrix} \theta_{ij} - \omega_j \text{ if } i \neq j \\ \theta_{ii} \text{ else} \end{pmatrix}_{ij} \end{aligned} \tag{s1}$$

maps an oMHN to its corresponding cMHN. Let $p^o(D \mid \cdot)$ and $p^c(D \mid \cdot)$ describe the oMHN and cMHN likelihood function, respectively, given some dataset D. Then $p^o(D \mid \cdot) = p^c(D \mid c(\cdot))$ and

$$\frac{\partial p^o(D \mid \cdot)}{\partial \theta_{ij}} = \frac{\partial p^c(D \mid \cdot)}{\partial \theta_{ij}} \circ c \qquad \text{and} \tag{s2}$$

$$\frac{\partial p^o(D \mid \cdot)}{\partial \omega_j} = -\sum_{i\neq j} \frac{\partial p^c(D \mid \cdot)}{\partial \theta_{ij}} \circ c\,. \tag{s3}$$

Now, if $\mathcal{I}^o(\cdot)$ and $\mathcal{I}^c(\cdot)$ are the Fisher information matrices for oMHN and cMHN, respectively, we have

$$(\mathcal{I}^o(\cdot))_{\theta_{ij}\theta_{kl}} = (\mathcal{I}^c(c(\cdot)))_{\theta_{ij}\theta_{kl}} \tag{s4}$$

$$(\mathcal{I}^o(\cdot))_{\omega_j\theta_{kl}} = -\sum_{i\neq j} (\mathcal{I}^c(c(\cdot)))_{\theta_{ij}\theta_{kl}} \qquad \text{and} \tag{s5}$$

$$(\mathcal{I}^o(\cdot))_{\omega_j\omega_l} = \sum_{i\neq j}\sum_{k\neq l} (\mathcal{I}^c(c(\cdot)))_{\theta_{ij}\theta_{kl}}\,. \tag{s6}$$

S1.1.2 Using the Metric Tensor

For smMALA, we need to use the metric tensor $G(\theta) = \mathcal{I}(\theta) - H(\log \mathrm{pr})$, with $\mathcal{I}(\theta)$ the Fisher information matrix of an MHN and $H(\log \mathrm{pr})$ the Hessian of the log-prior, in multiple ways:

$G(\theta)^{-1}$ As it is positive definite, the matrix $G(\theta)$ allows for a Cholesky decomposition $G(\theta) = LL^T$ with L a lower triangular matrix. Therefore we can efficiently compute $G(\theta)^{-1}x = L^{-T}L^{-1}x$ through substitution.

$\det G(\theta)$ Through its Cholesky decomposition we also get $G(\theta)$'s determinant efficiently as the squared product of L's diagonal entries.

drawing from $N(\mu, \epsilon G(\theta)^{-1})$ This can be achieved by drawing a multidimensional standard normal vector z and transforming it with $\sqrt{\epsilon}L^{-T}z + \mu$.

S1.2 Approximating the FIM

The Fisher information matrix can be approximated by drawing a sufficiently large set of samples $S \sim p(\cdot \mid \theta)$:

$$\mathcal{I}(\theta)_{ij} = E\left[\frac{\partial \log p}{\partial \theta_i}\frac{\partial \log p}{\partial \theta_j} \;\middle|\; \theta\right] \tag{s7}$$

$$\approx \frac{1}{|S|}\sum_{x\in S}\frac{\partial \log p}{\partial \theta_i}\frac{\partial \log p}{\partial \theta_j} \tag{s8}$$

In theory, this could be used to reduce the runtime of smMALA, as the biggest overhead is caused by the exact calculation of the Fisher information matrix. However, in our experiments, it took the MALA only few more steps to converge, the potential speed up would not be sufficient for smMALA to outperform MALA.

S1.3 Positive Definiteness of the Metric Tensor

For a matrix to induce a metric, it must be positive definite. A Fisher information matrix is always positive semi-definite, but we will show that it is in general not positive definite for MHN. In order for eq. (7) to be positive definite, the Hessian of the log-prior $H(\log \mathrm{pr})$ has to be negative definite. As we will show, this does not hold for all priors.

S1.3.1 Non-Positive Definiteness of the FIM for MHN

The Fisher information matrix $\mathcal{I}(\theta)$ is positive semi-definite but in general not positive definite: Vocht (2022) showed that, for a cMHN with $n = 2, 3, 4$, the FIM has a non-trivial nullspace and is therefore not positive definite. The FIM for oMHN is in fact never positive definite: This can be seen by realizing that infinitely many oMHNs have the same probability distribution. There is a subspace along which the model's parameters can be changed without its probability distribution changing.

Alternatively, one can see that the FIM has a non-trivial nullspace by multiplying it with

$$\begin{pmatrix} 0 & \text{if } i = j \\ 1 & \text{else} \end{pmatrix}_{\substack{i=1,\dots,n+1 \\ j=1,\dots,n}} . \tag{s9}$$

Therefore, it is never positive definite.

This reflects the non-identifiability of an unregularized MHN. Multiple distinct parameter configurations can produce the same probability distribution. This means that the distance induced by the FIM cannot serve as a metric, because it may assign a distance of zero to two points that are not the same.

S1.3.2 (Non-) Negative Definiteness of the Prior

To still use (7) as a metric tensor, this non-identifiability needs to be countered with a log-concave prior. The Hessian $H(\log \mathrm{pr})$ has to be negative definite (This corresponds to the prior pr being strictly log-concave).

This is neither true for the prior corresponding to a L1-penalty eq. (2) nor for the one corresponding to the symsparse penalty eq. (3): The log-Hessians of both have null-rows and columns, giving them a non-trivial nullspace. They are therefore not negative definite.

The log-Hessian of the prior corresponding to the L2-penalty is the identity matrix times some negative scalar and therefore negative definite. The log-Hessian H of the prior corresponding to the symmetrizing L2 penalty (4) is given by

$$H_{\vartheta_{ij}\vartheta_{kl}} = -\lambda \begin{cases} 2 & \text{if } i = k, j = l \\ -1 & \text{if } i = l, j = k, i \neq j \\ 0 & \text{else} \end{cases} \tag{s10}$$

and for any vector x we have

$$x^T H x = -\lambda \Big(\sum_i 2x_{\vartheta_{ii}}^2 + \sum_{i \neq j} (2x_{\vartheta_{ii}}^2 - x_{\vartheta_{ij}} x_{\vartheta_{ji}}) \Big) \tag{s11}$$

$$= -2\lambda \Big(\sum_i x_{\vartheta_{ii}}^2 + \sum_{i<j} (x_{\vartheta_{ii}}^2 - x_{\vartheta_{ij}} x_{\vartheta_{ji}} + x_{\vartheta_{jj}}^2) \Big). \tag{s12}$$

Every summand in the sums is non-negative and 0 if and only if every involved component of x is 0. Therefore H is negative definite.

S2 Pseudocode (sm)MALA

Here we show the pseudocode for the Metropolis-Adjusted Langevin Algorithm (MALA) and for simplified manifold MALA.

Algorithm SA1 MALA for MHN

Input: dataset D, prior π, initial distribution π_{init}, step size ϵ, number of iterations n_{it}
`samples` $\leftarrow []$
draw θ from π_{init}
`samples.append(`θ`)`
for $t = 1, \ldots, n_{it}$ **do**
  $\theta' \leftarrow$ draw from $q(\cdot \mid \theta) = N(\theta + \frac{\epsilon}{2} \frac{\partial(\log p(\theta|D) + \log \pi(\theta))}{\partial \theta}, \epsilon I)$
  $u \leftarrow$ draw from `uniform(0,1)`
  if $u < \frac{p(\theta'|D)\pi(\theta')q(\theta|\theta')}{p(\theta|D)\pi(\theta)q(\theta'|\theta)}$ **then**
    $\theta \leftarrow \theta'$
  end if
  `samples.append(`θ`)`
end for
Output: `samples`

Algorithm SA2 smMALA for MHN

Input: dataset D, prior π, initial distribution π_{init}, Fisher information matrix $\mathcal{I}$, Hessian of the log-prior H, step size ϵ, number of iterations n_{it}
`samples` $\leftarrow []$
draw θ from π_{init}
`samples.append(`θ`)`
for $t = 1, \ldots, n_{it}$ **do**
  $G(\theta) \leftarrow \mathcal{I}(\theta) - H(\theta)$ `# metric tensor`
  $\theta' \leftarrow$ draw from $q(\cdot \mid \theta) = N(\theta + \frac{\epsilon}{2} G(\theta)^{-1} \frac{\partial(\log p(\theta|D) + \log \pi(\theta))}{\partial \theta}, \epsilon G(\theta)^{-1})$
  $u \leftarrow$ draw from `uniform(0,1)`
  if $u < \frac{p(\theta'|D)\pi(\theta')q(\theta|\theta')}{p(\theta|D)\pi(\theta)q(\theta'|\theta)}$ **then**
    $\theta \leftarrow \theta'$
  end if
  `samples.append(`θ`)`
end for
Output: `samples`

S3 Mutation Risks

In this section we show the mutation risks for the remaining events as predicted by the MCMC samples.

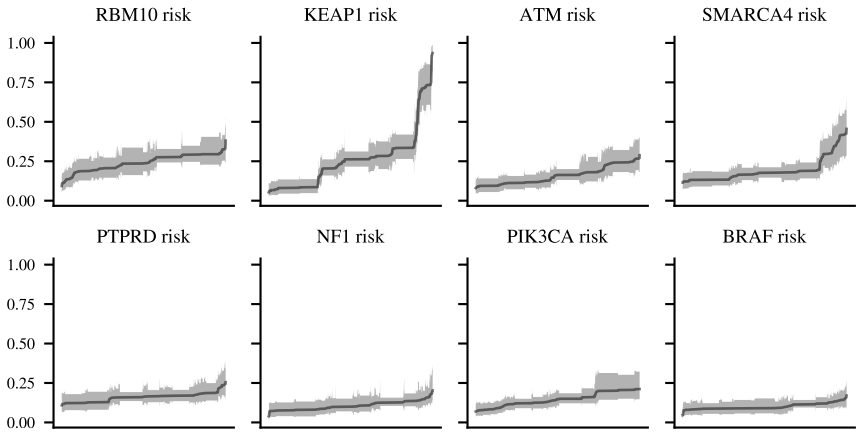

Figure SF1: Predicted mutation risks (mean and 90% posterior interval) across patients of the LUAD dataset, predicted by the MCMC samples. Only patients who do not exhibit this mutation at the time of sequencing are shown and patients are ordered by median risk. For every 100th MCMC sample, 1000 artificial tumors were sampled for 1 time unit in Markov time to infer a mutation risk.

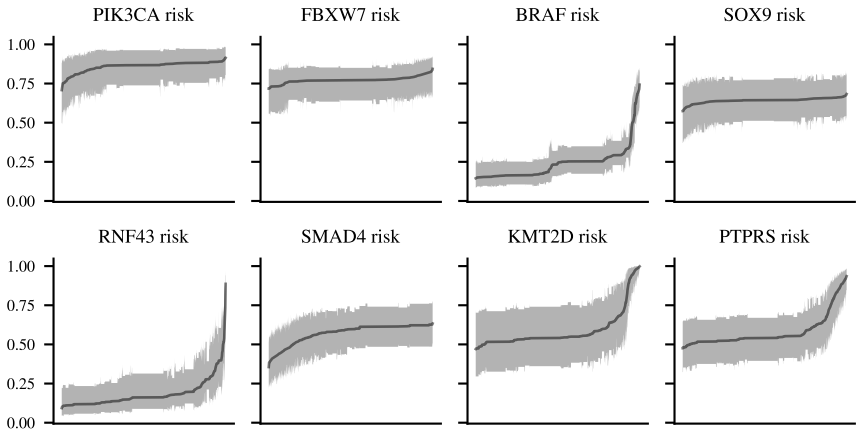

Figure SF2: Predicted mutation risks (mean and 90% posterior interval) across patients of the COAD dataset, predicted by the MCMC samples. Only patients who do not exhibit this mutation at the time of sequencing are shown and patients are ordered by median risk. For every 100th MCMC sample, 1000 artificial tumors were sampled for 1 time unit in Markov time to infer a mutation risk.

S4 Multimodal Posterior Distributions for the L1 MHNs

Here we show the posterior distributions for the LUAD and COAD L1 model, respectively.

S4.1 LUAD

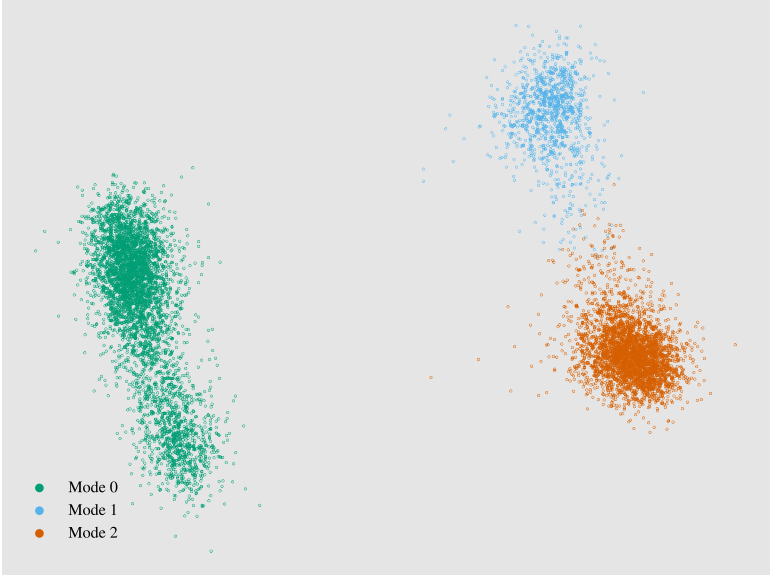

Figure SF3: First two principal components of posterior samples from an MHN trained on the LUAD dataset with a non-symmetrizing L1 penalty. Every 10th sample is shown and colored by cluster. Clusters were obtained by applying HDBSCAN to a 2-dimensional UMAPping.

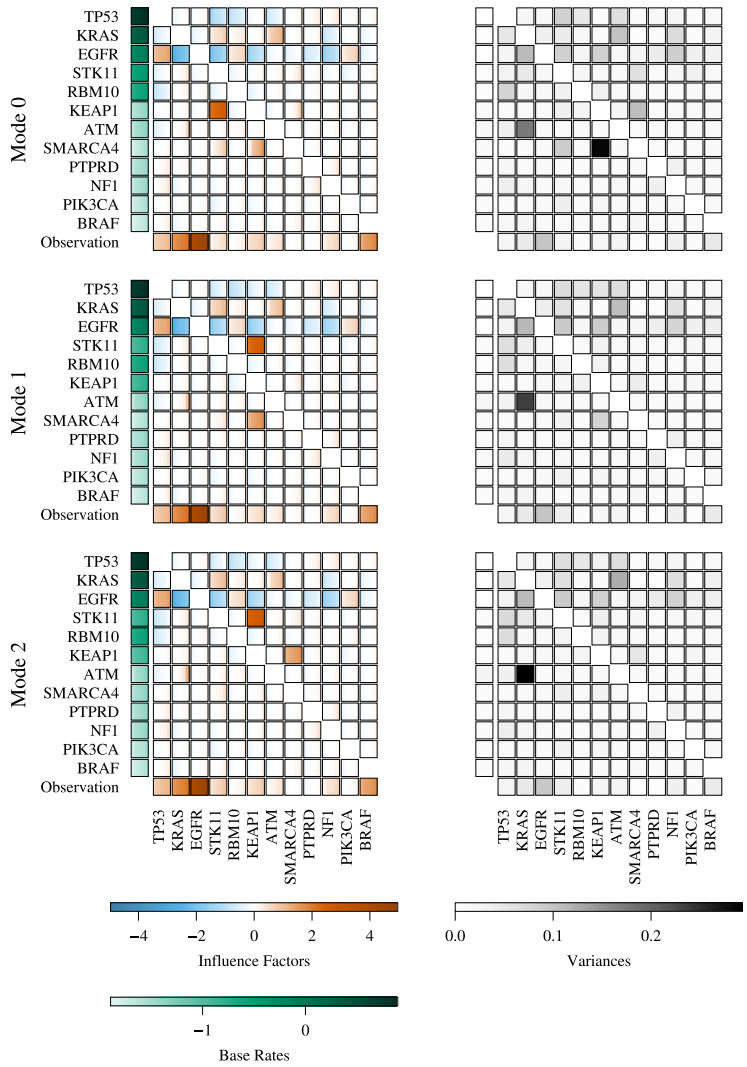

Figure SF4: Distribution and variance of the parameter matrix for the modes of the posterior samples of an MHN trained on the LUAD dataset with the L1 penalty (fig. SF3). Each cell in the left plots shows the 90% posterior interval of the corresponding parameter as a color gradient. E. g., the leftmost color in a cell corresponds to the 5% quantile of the parameter's distribution, the middle color to its 50% quantile, the rightmost color to its 95% quantile. Influence factors and observation rates are shown in blue/orange and base rates are shown in green. Each cell in the right plots shows the variance of the corresponding parameter. Note that for better visibility in the parameter matrices we are using different color scales than in main article.

S4.2 COAD

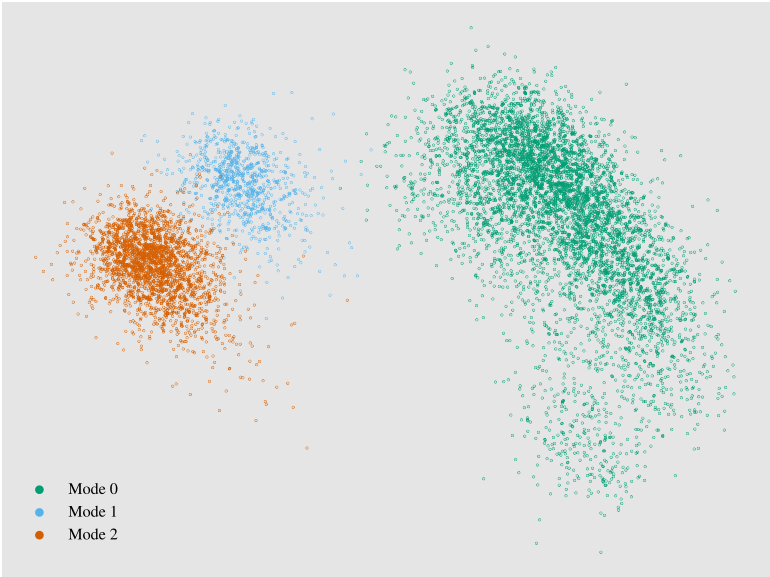

Figure SF5: First two principal components of posterior samples from an MHN trained on the COAD dataset with a non-symmetrizing L1 penalty. Every 10th sample is shown and colored by cluster. Clusters were obtained by applying HDBSCAN to a 2-dimensional UMAPping.

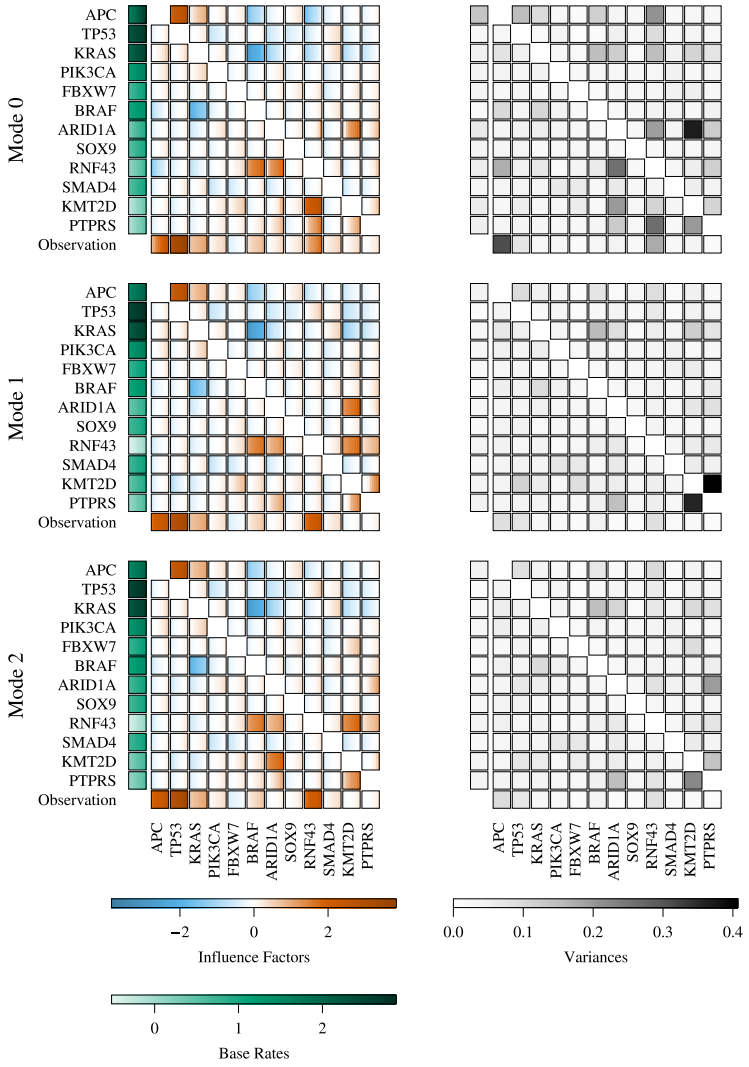

Figure SF6: Distribution and variance of the parameter matrix for the modes of the posterior samples of an MHN trained on the COAD dataset with the L1 penalty (fig. SF5). Each cell in the left plots shows the 90% posterior interval of the corresponding parameter as a color gradient. E. g., the leftmost color in a cell corresponds to the 5% quantile of the parameter's distribution, the middle color to its 50% quantile, the rightmost color to its 95% quantile. Influence factors and observation rates are shown in blue/orange and base rates are shown in green. Each cell in the right plots shows the variance of the corresponding parameter. Note that for better visibility in the parameter matrices we are using different color scales than in main article.

References

M. Girolami and B. Calderhead. Riemann manifold langevin and hamiltonian monte carlo methods. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 73(2):123–214, 03 2011. ISSN 1369-7412. doi: 10.1111/j.1467-9868.2010.00765.x.

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